FORTEBANK · DATATHON 26

CASHGUARD

Cashier video review: timecode, evidence frame and the NBRK clause — in minutes, no cloud

On-premise, open source · Under 5 minutes per video · A human makes the call

THE TASK

People watch hours of video. Show them only the seconds that matter

Verdict and evidence

"No violation found" or "Signs of possible violations", with type, timecode and 1–3 frames

Grounded in regulation

An NBRK clause from adilet.zan.kz, with no invented references or legal conclusions

Hard constraints

Local models only, keyed API, at most 5 minutes per video, a different camera angle in the final

> Per the case FAQ, a missed violation is unacceptable and false alarms should be as few as possible. The service is tuned for exactly that.

HOW IT WORKS

Two independent sources of findings

1 · DETECT

Frames and objects

Up to 4 frames/s, YOLOv8n and YOLO-World: people, phones, bags, cash

2 · PREPARE

Scene

The workplace is found automatically, the desk is zoomed in, CLIP embeddings twice a second

3 · JUDGE

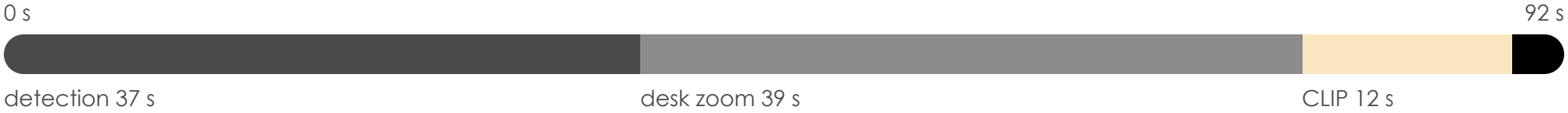
Rules + model

Rules explain down to the pixel, the model sees the whole scene. On overlap, the rule wins

> The model looks not at “which room is this” but at “what changed”: each embedding has the same video’s median subtracted. This helps on a new camera angle.

ONE VIDEO, END TO END

Video X-4, 73 seconds: report after 92 seconds



When	Where in code	What happens	Why
0 s	main.py, jobs.py	video to disk, job in PostgreSQL and the queue	API returns job_id at once
0–37 s	video.py, detector.py	291 frames, YOLOv8n and YOLO-World	frame rate fits the 5-minute limit
37–76 s	scene.py	workplace, desk ×2, small objects	a phone is 15 pixels wide
76–89 s	embedder.py, learned.py	CLIP twice a second, the model	see changes in the scene
89–92 s	rules.py, evidence.py, store.py	episodes, frames, clauses, storage	report and evidence

Found: “cash left open” at 50 s (label: 57 s), calibrated confidence 0.49 · CPU, 12 cores, no GPU

WHAT THE REVIEWER GETS

A moment, not the whole video

- Timecode with a jump to it in the player
- 1–3 frames with boxes around people and objects
- What is visible and why it may be a violation — kept separate
- The legal clause with a verbatim quote and link
- A “Human review” button: confirm or correct

Possible violation · 00:12.5

Unauthorised person in the cash room

What is visible: a person enters the cash room while valuables are being handled (2.5 s)

> NBRK Board Resolution No. 14 of 24.02.2020, para. 8: “The cash unit shall be isolated…”

A prompt for human review, not a legal conclusion

LEGAL BASIS

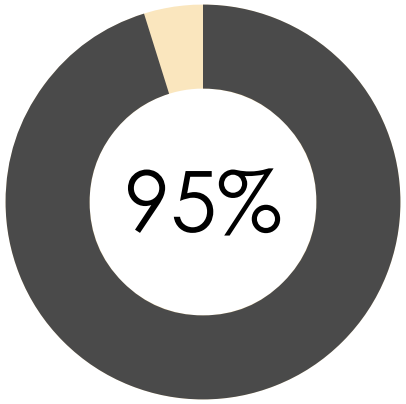
Two NBRK acts, references to direct clauses only

What is visible in the frame	Basis
Workplace, cash, bag or safe left open	No. 231, para. 20
Counting out of the client's view; client left before the count ended	No. 231, para. 31(4); para. 35
Counting without a controlling employee	No. 231, para. 42
Cash-collection bag accepted without checking	No. 231, para. 24(3)–(4); para. 25
Phone, personal items, cash into a pocket	No. 231, para. 12
Outsider in the cash room; cash-room door open; camera blocked	No. 14, para. 8; para. 16(1), (6)

No. 231 — Rules for cash operations and cash collection (29.11.2019). No. 14 — Rules for security and premises of second-tier banks (24.02.2020). Texts checked against the 2026 editions on adilet.zan.kz.

SERVICE RESULTS

Catches 21 of 22 violations on videos it has never seen



recall: 21 of 22 violations

15/20
false alarms on clean videos

4.5 s
median timecode error

Variant	Correct verdict	Recall	False alarms	MCC
Always “violation”	22/42	22/22	20/20	0.00
Rules only	20/42	4/22	4/20	−0.02
Trained model only	27/42	21/22	14/20	0.34
Rules + model (deployed)	**26/42**	**21/22**	**15/20**	**0.29**

42 labelled videos, 10 folds grouped by video: every number is on videos excluded from training

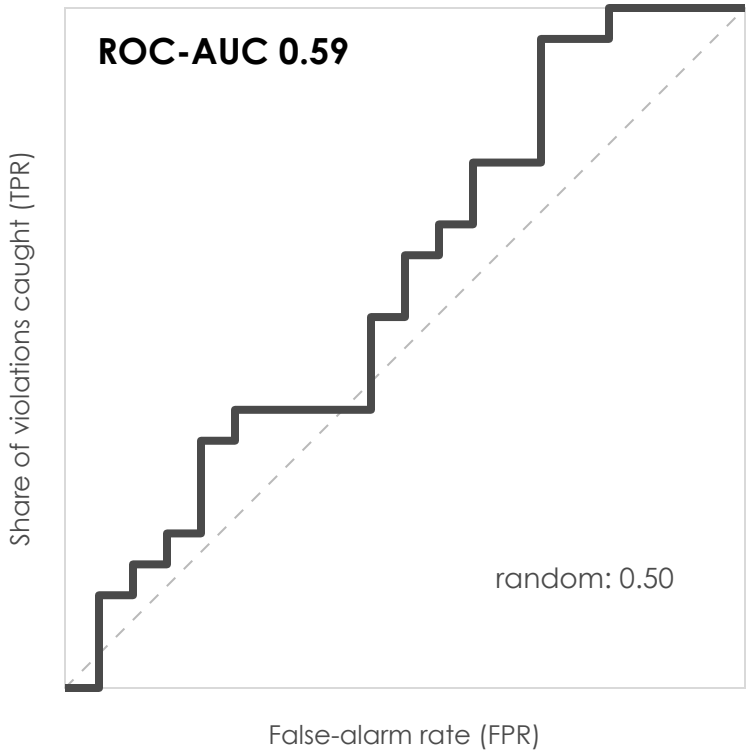
TRAINED MODEL

Metrics on held-out videos

Recall	Precision	F1
0.95	0.60	0.74

MCC	PR-AUC	Brier
0.34	0.60	0.30

	Flagged	Clean
Violation	21 TP	1 FN
No violation	14 FP	6 TN



CONFIDENCE AND THRESHOLD

The confidence in the report is a calibrated probability

Platt calibration

ECE before → ECE after

0.24 → 0.08

"70% confidence" now means a violation is present in roughly 7 cases out of 10. Calibration is applied only if it beats the raw score on held-out videos

The bank picks the threshold

Policy	Caught	False	Cost
MCC (current)	21/22	14/20	24
Minimum cost	**22/22**	**16/20**	**16**

Cost: miss = 10, false alarm = 1. Switched with a single setting, THRESHOLD_POLICY

ROBUSTNESS

Survives a new angle, not heavy compression

Each of the 42 videos was distorted and re-run through a model that had never seen it, at the production threshold.

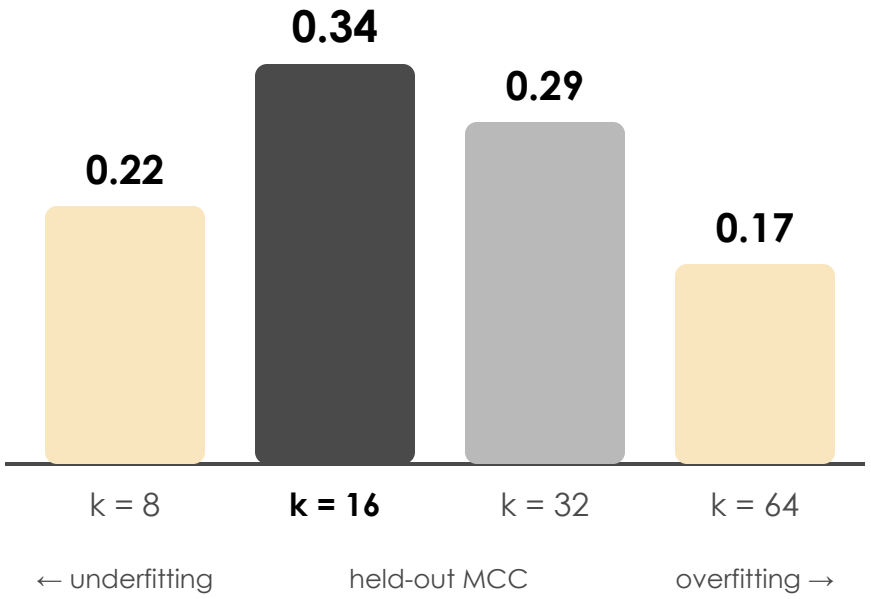
- Angle: ROC-AUC 0.60 vs 0.59 — subtracting the video's own "normal" works
- Darkness: recall drops to 0.73
- Heavy compression and occlusion: separation at chance level
- Augmentation was tested and rejected: it cuts recall on clean videos

Condition	Recall	False	ROC-AUC
No distortion	0.91	0.75	0.59
Different angle	**0.86**	**0.80**	**0.60**
Dark, ×0.35	0.73	0.55	0.63
Compressed, JPEG 12	0.86	0.90	0.50
Occluded 35%	0.86	0.80	0.46

OVER- AND UNDERFITTING

Held-out videos choose the model's complexity

- Cross-validation grouped by video: frames of one video never land in both training and test
- Grid: PCA size k and regularisation C
- The threshold is also chosen on held-out scores
- Every video carries equal weight
- ROC-AUC 0.76 on training, 0.61 held-out — a 0.15 gap, right at the line; the service shows this openly



CONTINUOUS LEARNING

The model learns from reviewers' confirmations

Report

Video embeddings are kept, the video itself is deleted

Reviewer

Violation or not, the second, the type — into the reviews table

Retraining

After 3 new answers, in the background, when the queue is empty

Gate

MCC no worse, recall not lower, no worse than the current model on new videos

Every version is kept

A row in model_versions: parameters, metrics, reason for the decision. Rolling back to any version is one click

Tested live

3 confirmations → v1 with MCC 0.38, but recall fell from 95% to 67%. The gate rejected it; the previous model kept running

TECHNICAL SOLUTION

Two containers, zero cloud calls

Service

FastAPI, job queue, React UI in RU / KZ / EN. API keyed with X-API-Key, as the brief requires

PostgreSQL 16

jobs, violations, evidence_frames, reviews, model_versions. CHECK constraints, one active model

Models

YOLOv8n, YOLO-World, CLIP ViT-B/32 — open source. Our model is 200 KB of numpy

171 s

longest video (142 s) on a CPU

0

network calls after the build

35

automated tests, including live Postgres

TEAM

Four areas of ownership

ML / CV

Aibar, Bekzhan

Detection, workplace inference, CLIP model, cross-validation, calibration, quality report

BACKEND / MLOPS

Abdulkarim, Aibar

API, queue and deadline, PostgreSQL, retraining and versions, Docker, tests

FRONTEND / UX

Abdulkarim

Upload, report and frames, human review, RU / KZ / EN, dark theme

DOMAIN / LEGAL

Daniyar

Labelling 42 videos, NBRK acts, texts and quotes, metrics, README, pitch

LIMITATIONS

What doesn’t work well yet, and why

Many false alarms

15 of 20 clean videos. A “correct exit” differs from a violation by a locked screen about 20 pixels wide

Type is often imprecise

22 violations across 13 situations: no exact type on new videos; in 10 of 22 cases a neighbouring type with the same clause is named

One room, compression

A different angle was only simulated. The model doesn’t yet withstand heavy compression or occlusion

If it isn’t visible, we don’t judge it

Postings, reconciliations, seals and authority are not checked from video. No face recognition, no audio

NEXT STEP

A pilot in one branch: 2–3 weeks of reviewer confirmations

Every confirmation is a new labelled example. The model retrains itself and goes live only if it is no worse

Keyed UI and API · Metrics — “Model quality” tab · Setup — README, 8 steps

THANK YOU